

# EQUALITY IN THE DEMAILLY–PHAM INEQUALITY AND MONOMIAL VALUATIONS

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**ABSTRACT.** We study equality in the Demailly–Pham inequality for plurisubharmonic germs in the local Cegrell class with positive Lelong number. We present a classification by diagonal logarithmic greenifications, whose coefficients are successive mixed-mass ratios. The proof combines strong openness, Zhou weights, hypersurface multiplicities, and weighted holomorphic extension. We also derive the finite-discrepancy case of Baily’s valuation conjecture in arbitrary codimension: an extremal real valuation is monomial in parameters of the original formal ring, and its center is regular. The coordinate lifting argument uses multiplier-ideal restriction and discrepancy rigidity.

## 1. INTRODUCTION

For a plurisubharmonic germ  $u$  at  $0 \in \mathbb{C}^n$ , its complex singularity exponent is

$$c_0(u) = \sup\{c > 0 : e^{-2cu} \in L_{\text{loc}}^1(0)\}.$$

We use the normalization  $dd^c = (i/\pi)\partial\bar{\partial}$ . When  $u$  belongs to the local Cegrell class  $\mathcal{E}$ , put

$$e_0(u) = 1, \quad e_j(u) = \nu((dd^c u)^j, 0) \quad (1 \leq j \leq n).$$

These are the classical masses retaining the contribution at the pole, not the non-pluripolar products.

If  $e_1(u) > 0$ , the inequality of Demailly and Pham [8] reads

$$(1.1) \quad c_0(u) \geq E_n(u) := \sum_{j=1}^n \frac{e_{j-1}(u)}{e_j(u)}.$$

The diagonal singularities  $\max_j a_j \log |z_j|$ , with  $0 < a_1 \leq \dots \leq a_n$ , attain equality. The question is whether these models describe all equality singularities.

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For ideals, this question is related to the study of diagonal ideals by Bivià-Ausina [3] and the homogeneous classification of Baily [1]. General plurisubharmonic germs require a different formulation: even an isolated equality singularity need not be a logarithmic model modulo a bounded function. Greenification removes this obstruction. Rashkovskii [13] proved the corresponding classification for the isotropic lower bound and studied the Demailly–Pham equality for multi-circled singularities.

Let  $D$  be a sufficiently small bounded hyperconvex neighborhood of 0, on which  $u$  has a negative representative. Its greenification is

$$(1.2) \quad G_{u,D} = (\sup\{q \in \text{PSH}^-(D) : q \leq u + O(1) \text{ near } 0\})^*.$$

We often write  $G_u$  when the domain is fixed.

**Theorem 1.1** (Equality classification). *Let  $u \in \mathcal{E}_{\text{loc}}$  near  $0 \in \mathbb{C}^n$ , and assume  $e_1(u) > 0$ . Then  $c_0(u) = E_n(u)$  if and only if there are holomorphic coordinates  $z_1, \dots, z_n$  centered at 0 such that*

$$(1.3) \quad G_u = \max_{1 \leq j \leq n} \left\{ \frac{e_j(u)}{e_{j-1}(u)} \log |z_j| \right\} + O(1).$$

*The assertion is local and holds for a sufficiently small choice of  $D$ . No analytic-singularity or rationality assumption is imposed.*

The classification concerns the greenification rather than the original germ modulo bounded functions. This distinction is necessary even for an isolated singularity; see [example 9.1](#). The coefficients are determined by the mixed masses, whereas the coordinate system need not be unique.

*Status of the manuscript.* The proof of [theorem 1.1](#) and the deductions below constitute a candidate argument awaiting independent mathematical verification. In particular, the valuation consequences are conditional on this analytic classification. This status applies to the manuscript as a whole and is separate from the published results cited in the proofs.

**1.1. The extension mechanism.** Suppose  $c = c_0(u) = E_n(u)$  and set  $\Phi = cG_u$ . The first part of the argument shows that  $\Phi$  is a Zhou weight with respect to 1. The results of Bao–Guan–Mi–Yuan [2] then supply a valuation

$$v(f) = \sigma(\log |f|, \Phi), \quad c_0^f(\Phi) = 1 + v(f),$$

where  $c_0^f$  denotes the exponent with multiplier  $|f|^2$ . For a sufficiently general linear equation  $h$  and  $H = (h = 0)$ , we establish

$$(1.4) \quad s := v(h) = \frac{e_{n-1}(\Phi)}{e_n(\Phi)}, \quad c_H := c_0(\Phi|_H) = E_{n-1}(\Phi|_H) = 1 - s.$$

The essential second statement is the exact lifting formula

$$(1.5) \quad \max_{F|_H=g} v(F) = \sigma(\log |g|, G_{\Phi|_H}), \quad 0 \neq g \in \mathcal{O}_{H,0}.$$

The maximum is over holomorphic germs, and is attained. Applying (1.5) to coordinates obtained in dimension  $n - 1$  produces coordinates in dimension

$n$  whose values sum to 1. The critical maximality of a diagonal Zhou weight finishes the proof.

The analytic inputs are strong openness, the Zhou weight theory, and the extension theorem with denominator  $|h|^2(-\log|h|)^2$ . The hypersurface mass calculation additionally uses Cutkosky's volume-multiplicity theorem [5]. No equality classification for general ideals, finite-generation theorem, or extraction of a divisor is used as an input. The approach is inspired by the critical integrability methods of Guan-Zhou [9, 10]; their Lelong-number-one classification is motivation, not a substitute for the induction below.

**1.2. A consequence for real valuations.** The classification has consequences for real valuations with finite log discrepancy. We first record the closed-point case, whose conclusion includes convergence of the monomializing parameters. Its proof is given in [section 7](#).

**Corollary 1.2** (Closed-point case of Baily's valuation conjecture). *Assume [theorem 1.1](#). Let  $R = \mathbb{C}[[x_1, \dots, x_n]]$ , and let  $v$  be a real valuation centered at its maximal ideal  $\mathfrak{m}$ , with  $A_R(v) < \infty$ . Put*

$$\mathfrak{a}_m(v) = \{f \in R : v(f) \geq m\}, \quad b_0(v) = 1, \quad b_j(v) = \lim_{m \rightarrow \infty} \frac{e_j(\mathfrak{a}_m(v))}{m^j}.$$

*If*

$$E_n(\mathfrak{a}_\bullet(v)) := \sum_{j=1}^n \frac{b_{j-1}(v)}{b_j(v)} = A_R(v),$$

*then there exist convergent holomorphic coordinates  $z_1, \dots, z_n$  at 0 such that, on the completed ring,*

$$v \left( \sum_{\alpha \in \mathbb{N}^n} c_\alpha z^\alpha \right) = \min_{c_\alpha \neq 0} \sum_{j=1}^n \alpha_j \frac{b_{j-1}(v)}{b_j(v)}.$$

*The value of the zero series is  $+\infty$ .*

Neither a divisorial nor a quasi-monomial hypothesis is imposed. The conclusion is monomiality in parameters on the original regular local ring, not merely on a birational model. Baily's Conjecture 6.3 in [1] also allows positive-dimensional centers and the truncated invariant  $E_l$ . The following extension is proved in [section 8](#).

**Corollary 1.3** (Arbitrary centers with finite discrepancy). *Assume [theorem 1.1](#). Let  $R = \mathbb{C}[[x_1, \dots, x_n]]$ , and let  $v$  be a real valuation centered at a prime  $\mathfrak{p}$  of height at least  $l \geq 1$ , with  $A = A_R(v) < \infty$ . If*

$$E_l(\mathfrak{a}_\bullet(v)) = A,$$

*then there is a regular system of parameters  $F_1, \dots, F_l, t_1, \dots, t_{n-l}$  of  $R$  in which  $v$  is monomial, with*

$$v(F_i) = \frac{b_{i-1}}{b_i} > 0, \quad v(t_j) = 0, \quad b_i = \lim_{m \rightarrow \infty} \frac{\sigma_i(\mathfrak{a}_m(v))}{m^i}, \quad b_0 = 1.$$

Here  $\sigma_i$  denotes the multiplicity on a general  $i$ -dimensional linear section. In particular,  $\mathbf{p} = (F_1, \dots, F_l)$  is regular and has height exactly  $l$ . The parameters are formal; their convergence is not asserted.

Thus the conditional consequence covers arbitrary centers and codimensions in the finite-discrepancy range. No assertion involving an extended-real equality  $E_l = A_R(v) = +\infty$  is made.

The regularity of the center in [corollary 1.3](#) is a conclusion, not a hypothesis. Restricting the valuation ideals to a general  $l$ -dimensional section produces an analytic equality germ. Multiplier ideals then lift its coordinates to the original ring with prescribed lower bounds on their values. Equality of the sum of these bounds with  $A_R(v)$  forces equality in the discrepancy inequality for the coordinate retraction. This last step identifies the original valuation and proves regularity of its center.

**1.3. Organization and dependencies.** Section [2](#) fixes the pluripotential inputs and proves preservation of all weighted exponents under greenification. Section [3](#) proves numerical rigidity, a finite radial truncation, and the Zhou-weight reduction, without induction. Section [4](#) relates valuation ideals to analytic masses. Section [5](#) proves adjunction and exact lifting; only then is induction used in Section [6](#). Section [7](#) constructs a psh equality germ from an arbitrary extremal valuation and recovers that same valuation from the diagonal model. Thus the valuation corollary is a consequence of the analytic classification and is not an input to its proof. Section [8](#) treats arbitrary centers by restricting the graded ideals to a general section, lifting its coordinates through multiplier ideals, and applying discrepancy rigidity on the original regular ring.

## 2. PLURIPOTENTIAL AND VALUATION PRELIMINARIES

**2.1. Comparison and greenification.** We use the local Cegrell class in the sense of [\[4\]](#). On a bounded hyperconvex domain  $D$ , the class  $\mathcal{E}_0(D)$  consists of bounded negative psh functions with zero boundary values and finite total Monge–Ampère mass. The class  $\mathcal{F}(D)$  consists of decreasing limits of functions in  $\mathcal{E}_0(D)$  with uniformly bounded total masses. A negative psh function belongs to  $\mathcal{E}(D)$  when it agrees locally with a function in  $\mathcal{F}(D)$  near every relatively compact subset. The notation  $\mathcal{E}_{\text{loc}}$  refers to membership on some neighborhood of the point under consideration. Mixed products are the classical Cegrell products. In particular, they retain their atoms at pluripolar points, and

$$e_j(u) = ((dd^c u)^j \wedge (dd^c \log |z|)^{n-j})(\{0\}).$$

Local finiteness and multilinearity hold for these products. The class is stable under passing to a less singular negative psh function. For  $u, v \in \mathcal{E}_{\text{loc}}$ ,

$$(2.1) \quad u \leq v + O(1) \implies e_j(u) \geq e_j(v), \quad c_0(u) \leq c_0(v).$$

*Proof of (2.1).* We include the singularity-comparison argument, since no pointwise order of the currents  $dd^c u$  and  $dd^c v$  is being asserted. After subtracting constants and shrinking to a common neighborhood, we may suppose  $u \leq v < 0$ . Constants do not affect any of the invariants under consideration.

First assume  $v(0) = -\infty$ . Fix  $0 < \varepsilon < 1$  and, for positive integers  $k$ , set

$$w_k = \max\{u, (1 - \varepsilon)v - k\}.$$

At points where  $v$  is finite,

$$(1 - \varepsilon)v - k - u \geq -\varepsilon v - k.$$

Upper semicontinuity and  $v(0) = -\infty$  imply that  $v(z) \rightarrow -\infty$  as  $z \rightarrow 0$ . For each fixed  $k$ , the right-hand side is therefore positive on a sufficiently small neighborhood of 0. Where  $v = -\infty$ , the order  $u \leq v$  also gives  $u = -\infty$ . Consequently, throughout that neighborhood,

$$w_k = (1 - \varepsilon)v - k, \quad e_j(w_k) = (1 - \varepsilon)^j e_j(v).$$

The neighborhood may depend on  $k$ , which is harmless for an invariant defined at the origin.

On the fixed common domain,  $w_k \downarrow u$  as  $k \rightarrow \infty$ , and  $w_k \geq u$  puts every  $w_k$  in  $\mathcal{E}_{\text{loc}}$ . The decreasing continuity of the Cegrell products [4] gives weak convergence of positive closed currents

$$(dd^c w_k)^j \rightharpoonup (dd^c u)^j.$$

Upper semicontinuity of Lelong numbers under weak convergence then yields

$$e_j(u) \geq \limsup_{k \rightarrow \infty} e_j(w_k) = (1 - \varepsilon)^j e_j(v).$$

Let  $\varepsilon \downarrow 0$  to obtain  $e_j(u) \geq e_j(v)$ . This is the truncation mechanism underlying Demailly's second comparison theorem; see [6, Section 5, Theorem 5.9].

For general  $v$ , work in a ball of radius less than 1 and put

$$u_\delta = u + \delta \log |z|, \quad v_\delta = v + \delta \log |z|, \quad \delta > 0.$$

The local Cegrell class is a convex cone, and  $\log |z|$  belongs to it, so  $u_\delta, v_\delta \in \mathcal{E}_{\text{loc}}$ . Moreover  $u_\delta \leq v_\delta$  and  $v_\delta(0) = -\infty$ . The preceding argument gives  $e_j(u_\delta) \geq e_j(v_\delta)$ . Multilinearity of the well-defined mixed products and linearity of the Lelong number give

$$e_j(u_\delta) = \sum_{\ell=0}^j \binom{j}{\ell} \delta^\ell \nu((dd^c u)^{j-\ell} \wedge (dd^c \log |z|)^\ell, 0) \rightarrow e_j(u).$$

The mixed Lelong numbers here are finite: equivalently they are point masses of the corresponding top-degree mixed Cegrell products after adjoining the remaining logarithmic factors. The same formula holds for  $v$ . Letting  $\delta \downarrow 0$  proves the first inequality in full generality.

Finally, if  $u \leq v + C$  and  $t > 0$ , then  $e^{-2tv} \leq e^{2tC} e^{-2tu}$ . Thus every integrable exponent for  $u$  is integrable for  $v$ , proving  $c_0(u) \leq c_0(v)$ .

Only weak convergence and upper semicontinuity were used above; continuity of the residual mass at 0 was not assumed.  $\square$

The positive mass vector is log-convex:

$$(2.2) \quad e_j(u)^2 \leq e_{j-1}(u)e_{j+1}(u), \quad 1 \leq j < n.$$

For the applications of these facts to (1.1), see [8, 13]. Rashkovskii's Lemma 1.1 and Lemma 2.1 in [13] give

$$(2.3) \quad \begin{aligned} G_u &\in \mathcal{F}(D), & (dd^c G_u)^n &= 0 \text{ on } D \setminus \{0\}, \\ c_0(G_u) &= c_0(u), & e_j(G_u) &= e_j(u), \end{aligned}$$

$$(2.4) \quad u \leq v + O(1), \quad e_n(u) = e_n(v) \implies G_u = G_v.$$

For isolated singularities bounded below by a logarithmic pole, the greenification is idempotent and homogeneous. These standard envelope properties are also obtained from the maximal Green-function description and uniqueness in  $\mathcal{F}(D)$ .

**2.2. Normalizations and local comparisons.** If  $a > 0$  and  $f$  is a nonzero holomorphic germ, our conventions give

$$(2.5) \quad e_j(aq) = a^j e_j(q), \quad c_0^f(aq) = a^{-1} c_0^f(q), \quad \sigma(p, a\Theta) = a^{-1} \sigma(p, \Theta).$$

All three follow directly from multilinearity or the defining integrability and comparison inequalities. Adding a bounded function to a weight changes none of these invariants. Constants in expressions of the form  $O(1)$  are local near 0; if an ideal index  $m$  is fixed, the constant is allowed to depend on  $m$ .

For a primary ideal  $I = (f_1, \dots, f_r)$  we write  $\log |I| = \frac{1}{2} \log \sum_i |f_i|^2$ . Another generating tuple changes this function by  $O(1)$ . With our normalization,  $e_j(\log |I|) = e_j(I)$  and  $c_0(\log |I|) = \text{let}(I)$ . The subscript  $H$  on an exponent or mass means that the invariant is computed in the smooth hypersurface  $H$ , using its own dimension. In particular,  $e_{n-1}(\Phi|_H)$  is a top mass on  $H$ , whereas  $e_{n-1}(\Phi)$  is a mixed mass in the ambient  $n$ -fold.

**2.3. The Zhou weight inputs.** We write

$$\begin{aligned} \mathcal{J}(q)_0 &= \{f \in \mathcal{O}_{\mathbb{C}^n, 0} : |f|^2 e^{-2q} \in L_{\text{loc}}^1(0)\}, \\ c_0^f(q) &= \sup\{t > 0 : f \in \mathcal{J}(tq)_0\}. \end{aligned}$$

Strong openness [9] implies nonintegrability at every finite critical exponent. For a maximal weight  $\Theta$ , the relative type is

$$\sigma(q, \Theta) = \sup\{b \geq 0 : q \leq b\Theta + O(1)\}.$$

**Definition 2.1.** A local Zhou weight with respect to 1 is a psh germ  $\Phi$  such that  $|z|^{2N} e^{-2\Phi}$  is integrable for some  $N$ ,  $e^{-2\Phi}$  is not integrable, and every psh germ  $\psi \geq \Phi + O(1)$  with  $e^{-2\psi}$  nonintegrable satisfies  $\psi = \Phi + O(1)$ . A global Zhou weight on  $D$  uses negative psh functions, pointwise order  $\psi \geq \Phi$ , and the conclusion  $\psi = \Phi$  on  $D$ .

We use the following published statements from [2]; the numbering refers to the journal version. Global Zhou weights induce local Zhou weights. They are maximal and locally bounded off their pole. Relative types to a Zhou weight are attained, and

$$(2.6) \quad v_\Phi(f) := \sigma(\log |f|, \Phi) \text{ is a valuation,} \quad c_0^f(\Phi) = 1 + v_\Phi(f)$$

for weights with respect to 1 (Corollary 1.9 and Theorem 1.10). In particular,

$$(2.7) \quad \mathcal{J}(t\Phi)_0 = \{f : v_\Phi(f) > t - 1\}.$$

We set  $v_\Phi(0) = +\infty$ . Remark 1.4 states that

$$(2.8) \quad q_k \longrightarrow q \text{ in } L_{\text{loc}}^1 \implies \limsup_k \sigma(q_k, \Theta) \leq \sigma(q, \Theta).$$

Theorem 1.11 identifies the order of relative types for all local Zhou weights with the corresponding inclusions of multiplier ideals. Finally, Example 1.5 says that

$$(2.9) \quad \max_i \frac{\log |z_i|}{w_i}, \quad w_i > 0, \quad \sum_i w_i = 1,$$

is a local Zhou weight with respect to 1.

**2.4. Multiplier ideals and increasing envelopes.** We isolate the limiting assertion needed both for greenification and for the passage from graded ideals to a psh germ.

**Lemma 2.2** (Increasing limits with a fixed holomorphic multiplier). *Let  $q_k$  be an increasing sequence of negative psh functions on a fixed neighborhood  $D$  of 0. Suppose that  $q = (\sup_k q_k)^*$  is not identically  $-\infty$ . Then, for every  $\lambda > 0$ ,*

$$\mathcal{J}(\lambda q)_0 = \bigcup_k \mathcal{J}(\lambda q_k)_0.$$

*Proof.* The standard increasing-limit theorem gives  $q_k \rightarrow q$  almost everywhere and in  $L_{\text{loc}}^1(D)$ . In particular the convergence holds in Lebesgue measure on relatively compact subdomains. Since  $q_k \leq q$ , integrability comparison gives the inclusion from right to left.

For the reverse inclusion put  $J_k = \mathcal{J}(\lambda q_k)_0$ . These ideals increase and stabilize, since  $\mathcal{O}_{\mathbb{C}^n, 0}$  is Noetherian; write their union as  $J_\infty$ . Suppose that  $g \in \mathcal{J}(\lambda q)_0 \setminus J_\infty$ . By Krull intersection there is an integer  $N$  with  $g \notin J_\infty + \mathfrak{m}^N$ . Thus a linear functional  $\Lambda$  on  $\mathcal{O}_{\mathbb{C}^n, 0}/\mathfrak{m}^N$  can be chosen so that it vanishes on the image of  $J_\infty$  but  $\Lambda(g) \neq 0$ . Choose a small domain  $U \Subset D$  on which  $g$  is represented. If  $F$  is holomorphic on  $U$  and  $(F - g, 0) \in J_k$ , then  $\Lambda(F) = \Lambda(g)$ . Cauchy estimates for derivatives at the interior point 0 give

$$|\Lambda(F)|^2 \leq C_{U, N, \Lambda} \int_U |F|^2.$$

Consequently

$$\inf_k \inf_{\substack{F \in \mathcal{O}(U) \\ (F-g, 0) \in J_k}} \int_U |F|^2 \geq \frac{|\Lambda(g)|^2}{C_{U,N,\Lambda}} > 0.$$

This bound holds on every sufficiently small such  $U$ . Apply the nonintegrability persistence theorem [2, Proposition 2.4] to the fixed multiplier  $g$ , auxiliary weight zero, and the weights  $\lambda q_k$ . It implies that  $|g|^2 e^{-2\lambda q}$  is not integrable at 0, contrary to the choice of  $g$ . This proves the equality. The argument also explains why upper-semicontinuous regularization of the increasing limit, which changes it only on a null set, does not affect the conclusion. The corresponding monotone statement is [2, Theorem 2.3].  $\square$

**Lemma 2.3** (Multiplier ideals under greenification). *For a negative psh function  $q$  and its greenification on  $D$ ,*

$$\mathcal{J}(tG_q)_0 = \mathcal{J}(tq)_0 \quad (t > 0).$$

Consequently  $c_0^f(G_q) = c_0^f(q)$  for every holomorphic germ  $f$ .

*Proof.* The function  $q$  belongs to the defining family in (1.2), and the family is stable under finite maxima. By Choquet's lemma, choose functions  $p_k$  in that family whose upper envelope has regularization  $G_q$ . Set  $q_k = \max\{q, p_1, \dots, p_k\}$ . For each fixed  $k$ ,

$$q \leq q_k \leq q + C_k$$

on some neighborhood of 0. The constant and the neighborhood may depend on  $k$ . Hence  $\mathcal{J}(tq_k)_0 = \mathcal{J}(tq)_0$  for every fixed  $k$  and  $t > 0$ . On the common domain  $D$  the sequence is increasing and its regularized limit is  $G_q$ . Apply lemma 2.2 to obtain

$$\mathcal{J}(tG_q)_0 = \bigcup_k \mathcal{J}(tq_k)_0 = \mathcal{J}(tq)_0.$$

Taking the supremum over the exponents for which a given holomorphic germ belongs to these ideals proves the final assertion.  $\square$

## 2.5. Diagonal logarithmic models.

**Lemma 2.4** (Masses, exponents, and coordinate types). *Let  $0 < a_1 \leq \dots \leq a_n$  and  $M_a = \max_i a_i \log |z_i|$  in a polydisk. Put  $w_i = a_i^{-1}$ . Then*

$$(2.10) \quad e_j(M_a) = a_1 \cdots a_j, \quad c_0^{z^\alpha}(M_a) = \sum_{i=1}^n (\alpha_i + 1) w_i \quad (\alpha \in \mathbb{N}^n).$$

Moreover  $\sigma(\log |z_i|, M_a) = w_i$ , and this relative type is attained. In particular  $c_0(M_a) = \sum_i w_i$  and  $E_n(M_a) = c_0(M_a)$ .

*Proof.* For sufficiently large  $t$ , the sublevel set  $\{M_a < -t\}$  is the polydisk with radii  $e^{-w_i t}$ . Direct integration gives

$$\int_{\{M_a < -t\}} |z^\alpha|^2 dV = C_\alpha \exp \left( -2t \sum_i (\alpha_i + 1) w_i \right).$$

Using  $e^{2\lambda s} = 1 + 2\lambda \int_0^s e^{2\lambda t} dt$  with  $s = -M_a$  and applying Tonelli's theorem shows that the weighted integral is finite exactly when  $\lambda < \sum_i (\alpha_i + 1)w_i$ . This includes divergence at the endpoint and proves the exponent formula.

For the mass formula first suppose all  $a_i$  are rational. Choose  $q \in \mathbb{N}$  with  $d_i = qa_i \in \mathbb{N}$ . The function  $M_a$  equals  $q^{-1} \log |(z_1^{d_1}, \dots, z_n^{d_n})| + O(1)$ . On a general  $j$ -dimensional linear section the restrictions of the first  $j$  coordinates are independent, and every later coordinate restricts to a linear combination of them. Every monomial in the restriction of  $z_k^{d_k}$  for  $k > j$  has total degree  $d_k \geq d_j$ , hence has weighted degree

$$\sum_{i=1}^j \frac{\alpha_i}{d_i} \geq \frac{d_k}{d_j} \geq 1.$$

The monomial integral-closure criterion therefore places that restriction in the integral closure of  $(z_1^{d_1}, \dots, z_j^{d_j})$  on the section. This diagonal complete intersection has multiplicity  $d_1 \cdots d_j$ . General-section multiplicities and invariance under integral closure, as recalled in [11, Chapter 17], give  $e_j(M_a) = q^{-j} d_1 \cdots d_j$ .

For real coefficients choose positive ordered rational  $a_i^- \leq a_i \leq a_i^+$  with both endpoints tending to  $a_i$ . On the unit polydisk,  $M_{a^+} \leq M_a \leq M_{a^-}$ . Mass comparison gives

$$a_1^- \cdots a_j^- \leq e_j(M_a) \leq a_1^+ \cdots a_j^+.$$

Pass to the limit to obtain the stated formula.

Finally,  $\log |z_i| \leq w_i M_a$  gives the lower bound on relative type. On the radial path  $|z_j| = e^{-w_j t}$  for all  $j$ ,  $M_a = -t$  and  $\log |z_i| = -w_i t$ . An inequality  $\log |z_i| \leq b M_a + O(1)$  with  $b > w_i$  fails on this path as  $t \rightarrow \infty$ . This proves the reverse bound, without requiring the weights to be rational.  $\square$

### 3. EQUALITY PRODUCES A CRITICAL ZHOU WEIGHT

**Lemma 3.1** (Strict numerical monotonicity). *Let  $p_0 = q_0 = 1$  and let  $p, q$  be positive log-convex vectors of length  $n + 1$ , with  $q_j \leq p_j$ . For  $\mathcal{D}$  the region of such vectors, put  $\mathcal{S}(t) = \sum_{j=1}^n t_{j-1}/t_j$ . Then*

$$\mathcal{S}(q) \geq \mathcal{S}(p), \quad \mathcal{S}(q) = \mathcal{S}(p) \implies q = p.$$

*Proof.* The region  $\mathcal{D}$  is convex. Indeed, for  $z_j = (1 - \tau)q_j + \tau p_j$ , log-convexity and Cauchy-Schwarz give

$$z_j \leq (1 - \tau)\sqrt{q_{j-1}q_{j+1}} + \tau\sqrt{p_{j-1}p_{j+1}} \leq \sqrt{z_{j-1}z_{j+1}}.$$

On this region,

$$\partial_j \mathcal{S} = -\frac{t_{j-1}}{t_j^2} + \frac{1}{t_{j+1}} \leq 0 \quad (j < n), \quad \partial_n \mathcal{S} = -\frac{t_{n-1}}{t_n^2} < 0.$$

Since  $p_j - q_j \geq 0$ , the chain rule gives

$$\frac{d}{d\tau} \mathcal{S}(q + \tau(p - q)) = \sum_{j=1}^n (p_j - q_j) \partial_j \mathcal{S}(q + \tau(p - q)) \leq 0.$$

Integrating proves monotonicity. If the endpoint values agree, this continuous nonpositive derivative has integral zero and hence vanishes at every  $\tau \in [0, 1]$ . Each summand is nonpositive, so each summand with  $p_j > q_j$  must vanish separately. Let  $r$  be the largest index with  $p_r > q_r$ . The last derivative rules out  $r = n$ . If  $r < n$ , vanishing forces  $z_r(\tau)^2 = z_{r-1}(\tau)z_{r+1}(\tau)$  throughout the segment. The right side is affine because  $z_{r+1}$  is constant, whereas the left side has positive quadratic coefficient  $(p_r - q_r)^2$ . This is impossible.  $\square$

**Proposition 3.2** (Ordered rigidity). *Assume  $u \in \mathcal{E}_{\text{loc}}$ ,  $e_1(u) > 0$ , and  $c_0(u) = E_n(u)$ . If  $u \leq q + O(1)$  and  $c_0(q) = c_0(u)$ , then  $e_j(q) = e_j(u)$  for every  $j$ , and  $G_q = G_u$ .*

*Proof.* Upward stability puts  $q$  in  $\mathcal{E}_{\text{loc}}$ . Its finite threshold implies  $e_1(q) > 0$  by Skoda's integrability criterion. Comparison, [lemma 3.1](#), and [\(1.1\)](#) give

$$c_0(u) = E_n(u) \leq E_n(q) \leq c_0(q) = c_0(u).$$

Apply the strict part of [lemma 3.1](#) with  $p_j = e_j(u)$  and  $q_j = e_j(q)$ . It identifies every mass, including the top mass. The two germs are ordered by hypothesis, so [\(2.4\)](#) now identifies their greenifications. The ordered hypothesis is essential here; equality of two numerical mass vectors alone would not identify unrelated singularities.  $\square$

**Proposition 3.3** (Finite logarithmic truncation). *Under the assumptions of [proposition 3.2](#), there is a finite  $T_0 > 0$  such that, for all  $T \geq T_0$ ,*

$$u_T = \max\{u, T \log |z|\}$$

*has the same exponent, mixed masses, and greenification as  $u$ . In particular  $G_u \geq T_0 \log |z| + O(1)$  near 0.*

*Proof.* Write  $c = c_0(u)$ . The ideal  $I = \mathcal{J}(cu)_0$  is proper by strong openness. Its zero set is isolated at 0. To see this, at every point  $x$  where  $\mathcal{J}(cu)_x$  is proper, Skoda's criterion gives  $\nu(u, x) \geq 1/c$ . The classical atom comparison gives

$$(dd^c u)^n \{x\} \geq \nu(u, x)^n \geq c^{-n}.$$

Local finiteness of the Cegrell Monge–Ampère measure allows only finitely many such points in a compact neighborhood. After shrinking, the zero set is  $\{0\}$ , so  $\mathfrak{m}^K \subset I$  for some  $K$ .

Choose representatives of the multiplier ideal on this smaller neighborhood. Each monomial  $z^\alpha$  with  $|\alpha| = K$  belongs to  $I$ , hence  $|z^\alpha|^2 e^{-2cu}$  is integrable near 0. Strong openness supplies a positive improvement for each of the finitely many monomials. Take their minimum and intersect their

neighborhoods. Since  $\sum_{|\alpha|=K} |z^\alpha|^2$  is comparable to  $|z|^{2K}$ , for some common  $\delta > 0$ ,

$$|z|^{2K} e^{-2(1+\delta)cu} \in L^1_{\text{loc}}(0).$$

Choose  $T_0$  with  $\delta c T_0 \geq K$  and restrict to  $|z| < 1$ . On  $B_T = \{u < T \log |z|\}$ , for  $T \geq T_0$ ,

$$e^{-2cu} = e^{2\delta cu} e^{-2(1+\delta)cu} \leq |z|^{2\delta c T} e^{-2(1+\delta)cu} \leq |z|^{2K} e^{-2(1+\delta)cu}.$$

Off  $B_T$  the two exponentials are equal, and on  $B_T$  their difference is nonnegative and bounded by the displayed integrable majorant. Thus  $e^{-2cu} - e^{-2cu_T}$  is integrable. If  $e^{-2cu_T}$  were integrable, their sum would make  $e^{-2cu}$  integrable, contrary to strong openness at the critical exponent. Since  $u_T \geq u$ , every subcritical exponent for  $u$  is integrable for  $u_T$ ; nonintegrability at  $c$  gives the opposite bound. Therefore  $c_0(u_T) = c$ , and [proposition 3.2](#) applies. A suitably shifted logarithmic pole is admissible in the greenification of  $u_T$ , which gives the last assertion. This is a radial version of the maximum construction in [\[10, Proposition 2.1\]](#).  $\square$

**Proposition 3.4.** *If  $c = c_0(u) = E_n(u)$ , then  $\Phi = cG_u$  is a global Zhou weight with respect to 1 on a sufficiently small  $D$ . Moreover*

$$(3.1) \quad L \log |z| + O(1) \leq \Phi \leq \mu \log |z| + O(1)$$

for finite  $L > 0$  and  $\mu = \nu(\Phi) > 0$ .

*Proof.* By [\(2.3\)](#) and homogeneity,  $c_0(\Phi) = E_n(\Phi) = 1$ . The logarithmic lower bound follows from [proposition 3.3](#); the upper bound follows from the usual radial maximum characterization of the Lelong number. The scaling identities [\(2.5\)](#) show explicitly that

$$E_n(cG_u) = c^{-1} E_n(G_u) = 1, \quad c_0(cG_u) = c^{-1} c_0(G_u) = 1.$$

In particular,  $\Phi$  is locally bounded off 0, and  $|z|^{2N} e^{-2\Phi}$  is integrable for large  $N$ . Strong openness gives nonintegrability of  $e^{-2\Phi}$ .

Let  $\psi \in \text{PSH}^-(D)$  satisfy  $\psi \geq \Phi$  and suppose  $e^{-2\psi}$  is not integrable at 0. Comparison gives  $c_0(\psi) \geq 1$ , while nonintegrability at exponent 1 gives  $c_0(\psi) \leq 1$ . Hence  $c_0(\psi) = 1$ . By [proposition 3.2](#),  $G_\psi = G_\Phi = \Phi$ . Since  $\Phi \leq \psi \leq G_\psi$ , it follows that  $\psi = \Phi$ . This is the global critical maximality condition. The local-global correspondence in [\[2\]](#) gives the local Zhou property.  $\square$

**Corollary 3.5.** *For an equality germ  $u$ , the standard Demailly approximants  $u_k$  satisfy  $e_j(u_k) \rightarrow e_j(u)$  for all  $1 \leq j \leq n$ .*

*Proof.* The approximation properties recalled in [\[8, Section 3.5\]](#) give  $e_j(u_k) \leq e_j(u)$ ,  $e_1(u_k) \rightarrow e_1(u)$ , and  $c_0(u_k) \rightarrow c_0(u)$ . Hence

$$E_n(u) \leq E_n(u_k) \leq c_0(u_k) \longrightarrow E_n(u).$$

Log-convexity bounds  $e_j(u_k)$  below by  $e_1(u_k)^j$ . Thus the mass vectors lie in a compact subset of the positive log-convex region. Every cluster point is

coordinatewise at most  $e(u)$  and has the same value of  $\mathcal{S}$ . By lemma 3.1 it equals  $e(u)$ .  $\square$

#### 4. VALUATION IDEALS AND HYPERSURFACE MASSES

Throughout this section  $\Phi$  is a local Zhou weight with respect to 1 satisfying (3.1). Set  $R = \mathcal{O}_{\mathbb{C}^n,0}$  and  $v = v_\Phi$ . The bounds imply

$$(4.1) \quad \frac{\text{ord}_0 f}{L} \leq v(f) \leq \frac{\text{ord}_0 f}{\mu} \quad (0 \neq f \in R).$$

For the lower inequality, use  $\log |f| \leq (\text{ord}_0 f) \log |z| + O(1)$  and the lower bound on  $\Phi$ . For the upper inequality, compare Lelong numbers in  $\log |f| \leq v(f)\Phi + O(1)$ . Define  $\mathfrak{a}_t = \{f : v(f) \geq t\}$  for  $t > 0$ , and  $\mathfrak{a}_t = R$  for  $t \leq 0$ . Then

$$(4.2) \quad \mathfrak{m}^{\lceil Lt \rceil} \subset \mathfrak{a}_t \subset \mathfrak{m}^{\lceil \mu t \rceil}.$$

The integer-indexed ideals form a graded family of primary ideals: the valuation inequalities make each  $\mathfrak{a}_t$  an ideal and give  $\mathfrak{a}_t \mathfrak{a}_r \subseteq \mathfrak{a}_{t+r}$ ; the first inclusion in (4.2) gives primaryness and finite colength. The second inclusion ensures a positive asymptotic volume.

**Lemma 4.1** (Singularity sandwich). *For  $m \in \mathbb{N}_{>0}$  put  $\phi_m = m^{-1} \log |\mathfrak{a}_m|$ , where the norm is taken on any finite generating tuple. Then*

$$(4.3) \quad (1 + m^{-1})\Phi + O(1) \leq \phi_m \leq \Phi + O(1).$$

Consequently, for every  $j$ ,

$$(4.4) \quad e_j(\Phi) \leq \frac{e_j(\mathfrak{a}_m)}{m^j} \leq (1 + m^{-1})^j e_j(\Phi).$$

*The same conclusions hold after restriction to any smooth hypersurface through 0, with the appropriate dimensional indices. Here the restricted ideal means its image in  $\mathcal{O}_{H,0}$ ; it is not being identified with a valuation ideal intrinsic to  $H$ .*

*Proof.* Each generator  $f$  of  $\mathfrak{a}_m$  satisfies  $\log |f| \leq v(f)\Phi + O(1) \leq m\Phi + O(1)$ , proving the upper bound after taking the norm of the finite generating tuple. Here  $\Phi$  is taken negative, and the largest of the finitely many additive constants suffices. By (2.7),  $\mathcal{J}((m+1)\Phi) = \{v > m\} \subset \mathfrak{a}_m$ . The Demailly approximation lower bound, in its local ideal form (see also the proof of [2, Lemma 3.9]), gives

$$(m+1)\Phi \leq \log |\mathcal{J}((m+1)\Phi)| + O(1) \leq \log |\mathfrak{a}_m| + O(1).$$

This proves (4.3). Classical comparison and the identity between ideal multiplicities and logarithmic masses give (4.4). More explicitly, the upper singularity bound gives  $e_j(\phi_m) \geq e_j(\Phi)$ , and the lower one gives  $e_j(\phi_m) \leq (1 + m^{-1})^j e_j(\Phi)$ ; also  $e_j(\phi_m) = m^{-j} e_j(\mathfrak{a}_m)$ . The constants may depend on  $m$ ; comparison is applied for each fixed  $m$  before taking a limit. A smooth

restriction has the same logarithmic bounds, hence an isolated pole and well-defined finite masses, so the same argument applies there. The image of any generating tuple generates the restricted ideal. Thus its logarithmic norm is the restriction of the original norm, and both singularity inequalities restrict with the same fixed-index constants. No uniformity in  $m$  is needed.  $\square$

We recall the precise algebraic input. If  $A$  is a regular local ring of positive dimension  $d$  and  $I_\bullet$  is a graded family of primary ideals, then [5, Theorem 6.5] gives

$$(4.5) \quad \lim_{m \rightarrow \infty} \frac{d! \ell(A/I_m)}{m^d} = \lim_{m \rightarrow \infty} \frac{e(I_m; A)}{m^d}.$$

Both limits exist. This applies to convergent power-series rings and their smooth hypersurface quotients.

**Proposition 4.2** (Hypersurface identity). *Assume  $n \geq 2$ . For a smooth hypersurface  $H = (h = 0)$  through 0,*

$$(4.6) \quad e_{n-1}(\Phi|_H) = v(h)e_n(\Phi).$$

*On the left, the mass is computed intrinsically on  $H$ .*

*Proof.* Write  $s = v(h)$  and introduce real-parameter length functions

$$P(t) = \ell(R/\mathfrak{a}_t), \quad Q(t) = \ell(R/(\mathfrak{a}_t + (h))).$$

Multiplicativity of  $v$  gives  $(\mathfrak{a}_t : h) = \mathfrak{a}_{t-s}$ . Multiplication by  $h$  therefore yields

$$(4.7) \quad 0 \longrightarrow R/\mathfrak{a}_{t-s} \xrightarrow{h} R/\mathfrak{a}_t \longrightarrow R/(\mathfrak{a}_t + (h)) \longrightarrow 0.$$

Thus  $Q(t) = P(t) - P(t-s)$ .

Apply (4.5) on  $R$  and  $R/(h)$  and use lemma 4.1. For integer  $m \rightarrow \infty$ ,

$$P(m) = \frac{V}{n!}m^n + o(m^n), \quad Q(m) = \frac{B}{(n-1)!}m^{n-1} + o(m^{n-1}),$$

where  $V = e_n(\Phi)$  and  $B = e_{n-1}(\Phi|_H)$ . Both  $P$  and  $Q$  are nondecreasing. For instance,  $P(\lfloor t \rfloor) \leq P(t) \leq P(\lceil t \rceil)$ , and the same holds for  $Q$ . Since  $\lfloor t \rfloor/t$  and  $\lceil t \rceil/t$  tend to 1, these inequalities extend the asymptotics to real parameters. The restricted ideals are graded and primary by (4.2), so the application of (4.5) on  $R/(h)$  is legitimate independently of whether they come from a valuation on that quotient. Since  $\mathfrak{a}_0 = R$ , we have  $P(0) = 0$ . Telescoping (4.7) along the step  $s$  gives

$$P(ks) = \sum_{i=1}^k Q(is) = \frac{Bs^{n-1}}{(n-1)!} \sum_{i=1}^k i^{n-1} + o(k^n) = \frac{Bs^{n-1}}{n!} k^n + o(k^n).$$

To justify the summed error, write the error in  $Q(is)$  as  $r_i = o(i^{n-1})$ . Given  $\varepsilon > 0$ , choose  $i_0$  so that  $|r_i| \leq \varepsilon i^{n-1}$  for  $i \geq i_0$ . The initial finite sum is  $O(1)$ , and the remaining sum is at most  $\varepsilon O(k^n)$ . Thus  $\sum_{i \leq k} r_i = o(k^n)$ . Finally  $\sum_{i \leq k} i^{n-1} = k^n/n + O(k^{n-1})$ . Comparison with  $P(ks) = Vs^n k^n/n! + o(k^n)$

gives  $B = sV$ . In particular, the argument does not differentiate, or take a finite difference of, an uncontrolled asymptotic remainder.  $\square$

**Corollary 4.3** (A common general hyperplane). *There is a linear form  $h$  such that  $H = (h = 0)$  satisfies*

$$(4.8) \quad e_j(\Phi|_H) = e_j(\Phi) \quad (1 \leq j \leq n-1), \quad v(h) = v(\mathfrak{m}) = \frac{e_{n-1}(\Phi)}{e_n(\Phi)}.$$

*Proof.* For each  $m$ , choose  $h$  in the nonempty Zariski open set for which the general-section formulas for the mixed multiplicities of  $\mathfrak{a}_m$  hold; see [11, Chapter 17]. There are countably many such conditions. In a fixed Euclidean chart of the space of hyperplanes each nonempty Zariski open set is open and dense, so their intersection is nonempty by the Baire theorem. This is the meaning of a common general hyperplane here; a single Zariski open set need not work for every  $m$ . We can also exclude the proper linear subspace on which  $v(h) > v(\mathfrak{m})$ . Indeed, for  $\gamma = \min_i v(x_i) = v(\mathfrak{m})$ , the map from linear forms to  $\{v \geq \gamma\}/\{v > \gamma\}$  is linear and nonzero. Its kernel is exactly that exceptional subspace. For this choice,  $e_j(\mathfrak{a}_m \mathcal{O}_H) = e_j(\mathfrak{a}_m)$  for all  $j < n$  and all  $m$ . Take limits using (4.4) on both spaces, and then apply [proposition 4.2](#).  $\square$

## 5. EXTENSION, ADJUNCTION, AND EXACT LIFTING

**5.1. The weighted extension statement.** We use the following rank-one form of the Ohsawa–Takegoshi–Manivel theorem; it is stated for arbitrary psh weights in [7, Corollary 8.9, pp. 46–47]. Let  $H = (h = 0)$  be a smooth hypersurface in a small bounded pseudoconvex domain  $U$ , with  $|h| < e^{-1}$ . If  $\psi$  is psh and  $g$  is holomorphic on  $H \cap U$  with  $\int_{H \cap U} |g|^2 e^{-2\psi} < \infty$ , then there is an extension  $F$  of  $g$  with

$$(5.1) \quad \int_U \frac{|F|^2 e^{-2\psi}}{|h|^2 (-\log |h|)^2} < \infty.$$

The harmless smooth Jacobian factor in the intrinsic measure is bounded above and below after shrinking  $U$ . In our applications a holomorphic coordinate system makes  $h$  one coordinate and  $U$  can be taken to be a small polydisk. The weight  $\psi$  is defined on a neighborhood of its closure, while integrability of the germ  $g$  on  $H$  is arranged by shrinking  $U$ . Multiplying  $h$  by a nonzero constant enforces  $|h| < e^{-1}$  without changing  $v(h)$  or the hypersurface. For every  $0 < a < 1$ , this implies

$$(5.2) \quad |F|^2 |h|^{-2a} e^{-2\psi} \in L^1_{\text{loc}}(0),$$

because  $r^{2(1-a)}(-\log r)^2$  is bounded near  $r = 0$ .

**Lemma 5.1** (A concavity bound). *Let  $\Phi$  be as in [section 4](#), let  $h$  define a smooth hypersurface, and let  $F \neq 0$  be holomorphic. Set  $s = v(h)$  and*

$$T_F(\tau) = \sup\{b \geq 0 : |F|^2 |h|^{2\tau} e^{-2b\Phi} \in L^1_{\text{loc}}(0)\}.$$

For  $0 < a < 1$ ,

$$(5.3) \quad T_F(-a) \leq 1 + v(F) - as.$$

*Proof.* For integrable pairs  $(b_i, \tau_i)$ ,  $i = 0, 1$ , and  $0 < \lambda < 1$ , the integrand at the convex combination is

$$(|F|^2|h|^{2\tau_0}e^{-2b_0\Phi})^{1-\lambda}(|F|^2|h|^{2\tau_1}e^{-2b_1\Phi})^\lambda.$$

Hölder's inequality makes it integrable. Taking increasing sequences of subcritical  $b_i$  proves concavity of  $T_F$ ; integrability at its endpoint is not assumed. Formula (2.6) and multiplicativity give  $T_F(0) = 1 + v(F)$  and  $T_F(1) = 1 + v(F) + s$ . Since  $0 = (1+a)^{-1}(-a) + a(1+a)^{-1} \cdot 1$ , concavity gives

$$(1+a)T_F(0) \geq T_F(-a) + aT_F(1).$$

Rearrangement proves the assertion. The parameter  $-a$  is admissible since  $h$  is smooth and  $|F|^2|h|^{-2a}$  is locally integrable for  $a < 1$ . In fact a small positive  $b$  is also admissible: use the lower logarithmic bound on  $\Phi$ , take  $p > 1$  with  $pa < 1$ , and apply Hölder to  $|h|^{-2a}$  and  $|z|^{-2bL}$ . Finiteness follows from  $T_F(-a) \leq T_F(0)$  after  $|h| < 1$ . The inequality at the negative parameter is an upper bound; reversing its direction would invalidate the adjunction argument.  $\square$

**Proposition 5.2** (Equality descends). *Let  $c_0(\Phi) = E_n(\Phi) = 1$  and choose  $H = (h = 0)$  as in corollary 4.3. With  $\varphi = \Phi|_H$  and  $s = v(h)$ ,*

$$(5.4) \quad c_H := c_0(\varphi) = E_{n-1}(\varphi) = 1 - s > 0.$$

*Proof.* For  $0 < t < c_H$ , apply (5.1) to  $g = 1$  and  $\psi = t\Phi$ . The extension  $F$  is a unit at 0. Thus (5.2) and lemma 5.1 imply  $t \leq 1 - as$  for every  $0 < a < 1$ . Letting  $a \uparrow 1$  and  $t \uparrow c_H$  gives  $c_H \leq 1 - s$ . By (4.8) and DP in dimension  $n - 1$ ,

$$c_H \geq E_{n-1}(\varphi) = E_n(\Phi) - \frac{e_{n-1}(\Phi)}{e_n(\Phi)} = 1 - s.$$

For a one-dimensional section the same inequality is the usual reciprocal-Lelong-number formula. The logarithmic bounds restrict to  $H$ , so  $0 < c_H < \infty$  before this argument begins. Consequently it is legitimate to take  $t \uparrow c_H$ , and the result also proves  $s < 1$ . No lower-dimensional equality classification has been used to establish (5.4).  $\square$

**Lemma 5.3** (Finite-jet attainment). *Let a valuation  $v$  on  $R$  satisfy (4.1). For each integer  $d \geq 0$ , it takes only finitely many values on the nonzero germs of order at most  $d$ . In particular, for  $0 \neq g \in \mathcal{O}_{H,0}$ , the supremum of  $v(F)$  over  $F|_H = g$  is attained.*

*Proof.* Choose an integer  $K$  with  $K/L > d/\mu$  and  $K > d$ . If  $\text{ord}_0 F \leq d$ , then  $v(F) \leq d/\mu$  while its Taylor tail of order at least  $K$  has value at least  $K/L$ . Write  $F = P + R_K$ , where  $P$  is its Taylor polynomial of degree less than  $K$ . The two values  $v(F)$  and  $v(R_K)$  are distinct. The elementary valuation rule  $v(A+B) = \min\{v(A), v(B)\}$  for unequal values therefore gives

$v(P) = v(F)$ . On a finite-dimensional complex vector space a valuation takes only finitely many finite values: vectors with pairwise distinct values are linearly independent, since a sum with a unique least-valued nonzero term has that value. For a lift of  $g$ , restriction cannot lower ordinary vanishing order, so  $\text{ord}_0 F \leq d := \text{ord}_H g$ . All lift values consequently belong to this same finite set, and there is at least one holomorphic lift in coordinates  $(y, h)$ . Hence their supremum is one of the realized values. Taylor polynomials were used only to bound the set of possible values: they need not restrict to  $g$ . The maximizing object is an original holomorphic lift, not a truncated polynomial or a formal limit of lifts.  $\square$

**Theorem 5.4** (Exact lifting). *In the setting of [proposition 5.2](#), every  $0 \neq g \in \mathcal{O}_{H,0}$  satisfies*

$$(5.5) \quad \max_{F|_H=g} v(F) = c_0^g(\varphi) - c_H = \sigma(\log |g|, G_\varphi).$$

*Proof.* Fix  $0 < t < c_0^g(\varphi)$ . Apply (5.1) to  $g$  and  $t\Phi$ , obtaining a holomorphic lift  $F$ . For every  $0 < a < 1$ , (5.2) and [lemma 5.1](#) give

$$t \leq T_F(-a) \leq 1 + v(F) - as.$$

The same  $F$  works for all  $a$ . Letting  $a \uparrow 1$  and using (5.4) yields  $v(F) \geq t - c_H$ . Taking the supremum over lifts and then  $t \uparrow c_0^g(\varphi)$  proves one inequality.

Conversely, for any lift  $F$ , attainment of relative type gives  $\log |F| \leq v(F)\Phi + O(1)$ . Restricting to  $H$  yields  $\log |g| \leq v(F)\varphi + O(1)$ . For  $0 < b < c_H$ , this yields

$$|g|^2 e^{-2(b+v(F))\varphi} \leq C e^{-2b\varphi},$$

which is integrable. Letting  $b \uparrow c_H$  proves  $c_0^g(\varphi) \geq c_H + v(F)$ . All weighted exponents used here are finite: the upper logarithmic bound on  $\varphi$  and the finite order of  $g$  force divergence at large enough exponents (one may integrate on a cone where the initial homogeneous part of  $g$  is nonzero). Thus the supremum equals the middle expression in (5.5). By [lemma 5.3](#) it is a maximum; no compactness of the family of extensions is needed.

By [proposition 5.2](#),  $\varphi$  is an equality germ in dimension  $n - 1$ . The dimension-independent [proposition 3.4](#) makes  $\Theta = c_H G_\varphi$  a Zhou weight with respect to 1. By [lemma 2.3](#) and (2.6),

$$c_0^g(\varphi) = c_0^g(G_\varphi) = c_H(1 + \sigma(\log |g|, \Theta)) = c_H + \sigma(\log |g|, G_\varphi).$$

This proves the final equality. The argument uses the Zhou reduction in lower dimension, not the classification that will be proved by induction.  $\square$

## 6. COORDINATE CONSTRUCTION AND THE CLASSIFICATION

**Lemma 6.1** (Critical coordinate criterion). *Let  $\Phi$  be a local Zhou weight with respect to 1. If  $z_1, \dots, z_n$  are holomorphic coordinates, then  $\sum_i v_\Phi(z_i) \leq 1$ . If equality holds, then*

$$\Phi = \max_i \frac{\log |z_i|}{v_\Phi(z_i)} + O(1).$$

*Proof.* Put  $w_i = v_\Phi(z_i) > 0$  and  $\Psi = \max_i w_i^{-1} \log |z_i|$ . Attainment of relative type gives  $\Psi \leq \Phi + O(1)$ . Since  $c_0(\Psi) = \sum_i w_i$  and  $c_0(\Phi) = 1$ , threshold comparison proves the inequality. If the sum is 1,  $\Psi$  is a Zhou weight by (2.9). The inequalities  $\Phi \geq \Psi + O(1)$  and  $e^{-2\Phi} \notin L_{\text{loc}}^1$  invoke the maximality of  $\Psi$ , giving  $\Phi = \Psi + O(1)$ .  $\square$

*Proof of theorem 1.1.* We prove the forward implication by induction on  $n$ . In dimension one, greenification of a subharmonic germ of positive Lelong number  $a$  is  $a \log |z| + O(1)$ . This follows from (2.3), harmonicity away from the pole, and the one-dimensional Green-function description. Indeed  $dd^c G_u = a \delta_0$  near the origin, so  $G_u - a \log |z|$  has zero distributional Laplacian there. It is therefore harmonic across the origin and bounded on a smaller disk. The exponent is  $1/a$ , which is  $E_1(u)$ .

Suppose the statement is known in dimension  $n - 1$ , and let  $u$  be an equality germ in dimension  $n$ . By proposition 3.4,  $\Phi = c_0(u)G_u$  is a Zhou weight with respect to 1 and has logarithmic bounds. Choose  $H = (h = 0)$  as in corollary 4.3, and put  $s = v_\Phi(h)$  and  $\varphi = \Phi|_H$ . By proposition 5.2,  $c_H = E_{n-1}(\varphi) = 1 - s$ .

The induction hypothesis supplies coordinates  $y_1, \dots, y_{n-1}$  on  $H$  and positive real numbers  $b_i$  such that

$$G_\varphi = \max_{i < n} b_i \log |y_i| + O(1), \quad \sum_{i < n} b_i^{-1} = c_H.$$

By lemma 2.4, the relative type of  $\log |y_i|$  to this model is exactly  $b_i^{-1}$ , even when the coefficients are irrational. By theorem 5.4, choose holomorphic lifts  $Y_i|_H = y_i$  with  $v_\Phi(Y_i) = b_i^{-1}$ . Their tangential differentials are independent. If a linear combination of  $dY_i(0)$  and  $dh(0)$  vanishes, restriction to  $T_0 H$  makes every coefficient of  $dY_i(0)$  vanish, since the  $dy_i(0)$  form a basis of  $T_0^* H$ . The remaining coefficient of  $dh(0)$  then vanishes because  $H$  is smooth. The holomorphic inverse function theorem therefore makes  $(Y_1, \dots, Y_{n-1}, h)$  a genuine local coordinate system. Moreover

$$\sum_{i < n} v_\Phi(Y_i) + v_\Phi(h) = c_H + s = 1.$$

The critical coordinate criterion now gives

$$(6.1) \quad \Phi = \max\{b_1 \log |Y_1|, \dots, b_{n-1} \log |Y_{n-1}|, s^{-1} \log |h|\} + O(1).$$

Rescale by  $c_0(u)$  to obtain a diagonal model for  $G_u$ .

Reorder the coefficients  $a_1, \dots, a_n$  of the resulting diagonal model increasingly. By lemma 2.4 and (2.3),

$$e_j(u) = a_1 \cdots a_j, \quad a_j = \frac{e_j(u)}{e_{j-1}(u)}.$$

This proves (1.3). Conversely, if (1.3) holds, the same mass and integrability formulas give

$$c_0(u) = c_0(G_u) = \sum_j a_j^{-1} = \sum_j \frac{e_{j-1}(u)}{e_j(u)} = E_n(u).$$

□

*Remark 6.2* (Real coefficients and holomorphic coordinates). The induction constructs convergent holomorphic germs directly by extension and finite-jet attainment. It does not obtain formal coordinates first, and it does not require rational values of the valuation. The ordered coefficients in (1.3) are nondecreasing by log-convexity of the mixed masses.

## 7. GRADED IDEALS AND EXTREMAL VALUATIONS AT A CLOSED POINT

This section proves [corollary 1.2](#) from [theorem 1.1](#). We keep separate the construction of an analytic equality germ and the recovery of the original valuation. An equality model with the same numerical invariants would not, by itself, establish the asserted classification of valuations.

**7.1. Finite jets and asymptotic invariants.** For a regular local ring  $R$ , a real valuation means a valuation of its fraction field, trivial on  $\mathbb{C}$  and nonnegative on  $R$ . Only the zero element has value  $+\infty$ . Its center is the prime ideal  $\{f \in R : v(f) > 0\}$ . For an ideal  $I \neq 0$  we set  $v(I) = \min_i v(f_i)$  for any finite generating set; the valuation inequalities make this independent of that set. We use the extension of log discrepancy to real valuations in [12], obtained as the supremum of the discrepancies of monomial retractions on log-smooth models. In particular

$$\text{lct}_R(I) \leq \frac{A_R(v)}{v(I)}$$

whenever  $0 < v(I) < \infty$  and  $A_R(v) < \infty$ .

Write  $\widehat{R} = \mathbb{C}[[x_1, \dots, x_n]]$  and  $R_{\text{an}} = \mathbb{C}\{x_1, \dots, x_n\}$ . Let  $v$  be as in [corollary 1.2](#), and set  $A = A_{\widehat{R}}(v)$  and  $s = v(\mathfrak{m}) > 0$ . The finite-discrepancy Izumi estimate [12, Proposition 5.10] says that

$$(7.1) \quad s \text{ord}_0(f) \leq v(f) \leq A \text{ord}_0(f).$$

In particular,  $0 < A < \infty$ , and for every integer  $m \geq 1$ ,

$$(7.2) \quad \mathfrak{m}^{\lceil m/s \rceil} \subseteq \mathfrak{a}_m(v) \subseteq \mathfrak{m}^{\lceil m/A \rceil}.$$

**Lemma 7.1** (Passage to convergent ideals). *Put  $I_m = \mathfrak{a}_m(v) \cap R_{\text{an}}$ . Then  $I_m \widehat{R} = \mathfrak{a}_m(v)$ ; the ideals  $I_m$  are primary, form a graded family, and satisfy (7.2) in the convergent ring. They have the same mixed multiplicities and log canonical thresholds as their completions.*

*Proof.* Fix  $m$  and choose  $N = \lceil m/s \rceil$ . If  $f \in \mathfrak{a}_m(v)$  and  $p$  is its Taylor polynomial of degree less than  $N$ , then  $f - p \in \mathfrak{m}^N \subseteq \mathfrak{a}_m(v)$ , so  $p \in \mathfrak{a}_m(v)$ . The quotient  $\mathfrak{a}_m(v)/\mathfrak{m}^N$  is a finite-dimensional vector space. Choose a basis represented by such polynomials, and adjoin the finitely many monomials of degree  $N$ . These polynomials generate  $\mathfrak{a}_m(v)$  in  $\widehat{R}$  and belong to  $I_m$ . They prove the extension assertion. Faithful flatness of completion, or the same finite-jet argument, identifies its contraction with  $I_m$ . The graded property follows by contracting  $\mathfrak{a}_m(v)\mathfrak{a}_r(v) \subseteq \mathfrak{a}_{m+r}(v)$ . Mixed multiplicities are unchanged under completion since the relevant finite-length quotients and their Hilbert polynomials are unchanged. Invariance of  $\text{lct}$  under this regular completion map follows from [12, Proposition 1.9].  $\square$

For each  $1 \leq j \leq n$ , choose a single general  $j$ -dimensional linear subspace that computes  $e_j(I_m)$  for every  $m$ . Such a choice exists by the countable-intersection argument in [corollary 4.3](#). On its convergent local ring, the restricted ideals are graded and primary, with the same power bounds as in (7.2). The volume–multiplicity theorem (4.5) therefore shows that

$$(7.3) \quad b_j = \lim_{m \rightarrow \infty} \frac{e_j(I_m)}{m^j}$$

exists. The power bounds and multiplicity comparison give

$$A^{-j} \leq b_j \leq s^{-j}.$$

The vector  $(b_0, b_1, \dots, b_n)$ , with  $b_0 = 1$ , is log-convex, by passage to the limit from the mixed multiplicities of  $I_m$ . Its last entry is the valuation volume:

$$b_n = \text{vol}(v) := \lim_{m \rightarrow \infty} \frac{n! \ell(\widehat{R}/\mathfrak{a}_m(v))}{m^n}.$$

The asymptotic threshold is  $\text{lct}(I_\bullet) = \sup_{m \geq 1} m \text{lct}(I_m)$ . For completeness, the two bounds needed here are

$$(7.4) \quad \sum_{j=1}^n \frac{b_{j-1}}{b_j} \leq \text{lct}(I_\bullet) \leq A.$$

For the first, apply DP to  $m^{-1} \log |I_m|$  and take limits: its  $j$ th mass is  $m^{-j} e_j(I_m)$  and its threshold is  $m \text{lct}(I_m)$ . For the second, the valutive bound for  $\text{lct}$  on  $\widehat{R}$  gives

$$m \text{lct}(I_m) = m \text{lct}(\mathfrak{a}_m(v)) \leq \frac{mA}{v(\mathfrak{a}_m(v))} \leq A.$$

Here  $v(\mathfrak{a}_m(v)) \geq m$ , and the valutive bound follows from the discrepancy description in [12]. Under the equality hypothesis of [corollary 1.2](#),

$$(7.5) \quad \text{lct}(I_\bullet) = A.$$

**7.2. An increasing Green-function realization.** We now formulate the construction for a general graded family, so that it can later be applied to the images of valuation ideals on a section. Let  $I_\bullet$  be a graded family of primary ideals in  $\mathbb{C}\{x_1, \dots, x_n\}$  satisfying, for some  $0 < \rho \leq C < \infty$ ,

$$(7.6) \quad \mathfrak{m}^{\lceil Cm \rceil} \subseteq I_m \subseteq \mathfrak{m}^{\lceil \rho m \rceil} \quad (m \geq 1).$$

Common general sections and (4.5) give the limits  $b_j = \lim_m m^{-j} e_j(I_m)$ , with  $\rho^j \leq b_j \leq C^j$  and  $b_0 = 1$ . Assume

$$(7.7) \quad \text{lct}(I_\bullet) = \sum_{j=1}^n \frac{b_{j-1}}{b_j} =: A.$$

These hypotheses have just been verified for the contracted valuation ideals under the equality hypothesis of [corollary 1.2](#), with  $\rho = A_R(v)^{-1}$  and  $C = v(\mathfrak{m})^{-1}$ . In this subsection  $A$  denotes the common finite number in (7.7); the construction does not require a valuation giving the ideals  $I_m$ .

Fix a small bounded hyperconvex domain  $D$  centered at 0, and let  $g_D = G_{\log|x|, D}$  be its ordinary logarithmic Green function. For a primary ideal  $I$ , write  $G_{I,D} = G_{\log|I|, D}$ . These envelopes depend only on the germ of the ideal at 0, so they can all be considered on this same domain. Ideal inclusion and homogeneity give

$$I \subseteq J \implies G_{I,D} \leq G_{J,D}, \quad G_{Ir,D} = rG_{I,D}.$$

Indeed,  $\log|I| \leq \log|J| + O(1)$  implies inclusion of the defining envelope families. Homogeneity follows by dividing every admissible negative psh function by  $r$ .

Set

$$(7.8) \quad q_k = \frac{1}{k!} G_{I_{k!}, D}, \quad q = \left( \sup_k q_k \right)^*.$$

Since  $I_{k!}^{k+1} \subseteq I_{(k+1)!}$ , we have  $q_k \leq q_{k+1}$ . The uniform bounds

$$(7.9) \quad (C+1)g_D \leq q_k \leq \rho g_D$$

follow from (7.6). Indeed, for  $m = k!$ ,

$$\frac{\lceil Cm \rceil}{m} g_D \leq \frac{1}{m} G_{I_m, D} \leq \frac{\lceil \rho m \rceil}{m} g_D.$$

Since  $g_D \leq 0$ ,  $\lceil Cm \rceil/m \leq C+1$ , and  $\lceil \rho m \rceil/m \geq \rho$ , the signs in (7.9) follow. Thus  $q_k \rightarrow q$  almost everywhere and in  $L_{\text{loc}}^1$ , and  $q$  is a negative psh function with an isolated pole, finite logarithmic lower bound, and positive Lelong number. In particular  $q \in \mathcal{E}_{\text{loc}}$ .

**Lemma 7.2** (Preservation of the critical threshold). *For the function in (7.8),  $c_0(q) = A$  under (7.7).*

*Proof.* Put  $d_m = m \text{lct}(I_m)$ . If  $m$  divides  $r$ , then  $I_m^{r/m} \subseteq I_r$  implies  $d_r \geq d_m$ . Consequently  $d_{k!}$  is increasing and tends to  $\sup_m d_m = \text{lct}(I_\bullet) = A$ : every

fixed  $m$  divides all sufficiently large factorials. By greenification invariance,  $c_0(q_k) = d_k!$ . Since  $q \geq q_k$ , threshold comparison gives  $c_0(q) \geq A$ .

We give the opposite inequality with the limiting integrability step made explicit. Fix  $t > A$ . For every  $k$ , the ideal  $J_k = \mathcal{J}(tq_k)_0$  is proper, since  $c_0(q_k) \leq A < t$ . On any sufficiently small fixed domain  $U$  containing 0, define

$$C_k(U) = \inf \left\{ \int_U |F|^2 : F \in \mathcal{O}(U), (F - 1, 0) \in J_k \right\}.$$

Because a proper ideal of the local ring is contained in  $\mathfrak{m}$ , every such  $F$  has  $F(0) = 1$ . The point-evaluation estimate on  $U$  gives  $\int_U |F|^2 \geq C_U > 0$ , where  $C_U$  is independent of  $k$ . Hence  $\inf_k C_k(U) > 0$ .

Now  $tq_k \rightarrow tq$  in Lebesgue measure. The nonintegrability persistence theorem [2, Proposition 2.4], applied with holomorphic multiplier 1 and auxiliary weight 0, implies  $e^{-2tq} \notin L_{\text{loc}}^1(0)$ . The numbering here is that of the journal version; the corresponding statement in arXiv v3 is Proposition 2.9. Since  $t > A$  was arbitrary,  $c_0(q) \leq A$ . Upper-semicontinuous regularization changes the increasing limit only on a null set, so it has no effect on this application.  $\square$

**Proposition 7.3** (Realization of extremal graded ideals). *Let  $I_\bullet$  satisfy (7.6) and (7.7). The function  $q$  in (7.8) belongs to  $\mathcal{E}_{\text{loc}}$  and satisfies*

$$e_j(q) = b_j \quad (1 \leq j \leq n), \quad c_0(q) = E_n(q) = A.$$

*Proof.* Since  $q \geq q_k$ , mass comparison gives

$$e_j(q) \leq e_j(q_k) = \frac{e_j(I_k!)}{(k!)^j}.$$

Taking the limit yields  $e_j(q) \leq b_j$ . Both vectors are positive and log-convex, by (7.9) and the limiting mixed-multiplicity inequalities. Apply numerical monotonicity and DP:

$$A = \sum_{j=1}^n \frac{b_{j-1}}{b_j} \leq E_n(q) \leq c_0(q) = A.$$

The strict assertion in lemma 3.1 now forces  $e_j(q) = b_j$  for each  $j$ . In particular, no general continuity assertion for residual Monge–Ampère masses under increasing limits is required. Here continuity of the masses is a consequence of extremality.  $\square$

**7.3. A hypersurface formula for arbitrary valuations.** The next observation does not involve Zhou weights. Let  $\eta$  be any centered real valuation on a regular local ring  $S$  of dimension  $n \geq 2$ , with  $c_1 \text{ ord} \leq \eta \leq c_2 \text{ ord}$  for constants  $c_1, c_2 > 0$ . Set  $\mathfrak{a}_t^\eta = \{\eta \geq t\}$  for  $t > 0$  and  $\mathfrak{a}_t^\eta = S$  for  $t \leq 0$ . For a smooth parameter  $h$  define

$$B_h(\eta) = \lim_{m \rightarrow \infty} \frac{e((\mathfrak{a}_m^\eta + (h))/(h); S/(h))}{m^{n-1}}.$$

The limit exists by (4.5), since the quotient ideals are graded and primary. The same theorem defines  $\text{vol}(\eta)$ .

**Lemma 7.4** (Valuative hypersurface volume). *With this notation,*

$$B_h(\eta) = \eta(h) \text{vol}(\eta).$$

*Proof.* Put  $a = \eta(h) > 0$ ,  $P_\eta(t) = \ell(S/\mathfrak{a}_t^\eta)$ , and  $Q_\eta(t) = \ell(S/(\mathfrak{a}_t^\eta + (h)))$ . The valuation equality  $\eta(hf) = a + \eta(f)$  gives the exact colon identity  $(\mathfrak{a}_t^\eta : h) = \mathfrak{a}_{t-a}^\eta$ . The resulting multiplication sequence gives  $Q_\eta(t) = P_\eta(t) - P_\eta(t-a)$ . By (4.5) in  $S$  and  $S/(h)$ , and integer-to-real monotone squeezing,

$$P_\eta(t) = \frac{\text{vol}(\eta)}{n!} t^n + o(t^n), \quad Q_\eta(t) = \frac{B_h(\eta)}{(n-1)!} t^{n-1} + o(t^{n-1}).$$

Sum the exact equality at  $t = a, 2a, \dots, ka$ :

$$\frac{\text{vol}(\eta)a^n}{n!} k^n + o(k^n) = P_\eta(ka) = \sum_{i=1}^k Q_\eta(ia) = \frac{B_h(\eta)a^{n-1}}{n!} k^n + o(k^n).$$

The control of the summed remainder is exactly the elementary  $\varepsilon$  estimate in the proof of [proposition 4.2](#). Comparison of leading coefficients proves the lemma.  $\square$

**Lemma 7.5** (Recovery from a dominating monomial valuation). *Suppose  $\eta$  and  $w$  are centered, linearly bounded real valuations on  $\mathbb{C}[[z_1, \dots, z_n]]$ ,  $w$  is monomial in these parameters,  $w \geq \eta$ , and  $\text{vol}(w) = \text{vol}(\eta) > 0$ . Then  $w = \eta$ . The same conclusion holds first on the convergent subring and then on its completion.*

*Proof.* Assume first  $n \geq 2$ . The order  $w \geq \eta$  means  $\mathfrak{a}_m^\eta \subseteq \mathfrak{a}_m^w$ . On each smooth quotient  $S/(z_j)$ , monotonicity of multiplicity therefore gives  $B_{z_j}(w) \leq B_{z_j}(\eta)$ . By [lemma 7.4](#) and equality of volumes,

$$w(z_j) \text{vol}(w) \leq \eta(z_j) \text{vol}(\eta),$$

so  $w(z_j) \leq \eta(z_j)$ . The assumed order gives the reverse inequality; hence equality holds for every parameter.

For a polynomial  $P = \sum_\alpha c_\alpha z^\alpha$ , the valuation inequality and these coordinate equalities show

$$\eta(P) \geq \min_{c_\alpha \neq 0} \sum_j \alpha_j \eta(z_j) = \min_{c_\alpha \neq 0} \sum_j \alpha_j w(z_j) = w(P).$$

Together with  $w \geq \eta$ , this proves equality on polynomials. For a nonzero convergent or formal series  $F$ , let  $P_N$  be its Taylor polynomial of degree less than  $N$ . Both valuations are bounded below by positive multiples of  $\text{ord}$ , so the values of  $F - P_N$  tend to  $+\infty$ . Their upper linear bounds ensure that their values on  $F$  are finite. For sufficiently large  $N$ , the unequal-value rule gives  $\eta(F) = \eta(P_N) = w(P_N) = w(F)$ . For  $n = 1$ , each centered real valuation is its value on the uniformizer times  $\text{ord}$ ; its volume is the reciprocal of that value, and the conclusion follows directly.  $\square$

#### 7.4. Completion of the proof of the valuation corollary.

*Proof of corollary 1.2.* By lemma 7.1 and proposition 7.3, construct an equality germ  $q$  with mass vector  $b$ . Apply the assumed theorem 1.1, shrinking the domain if needed, to obtain convergent coordinates  $z_j$  and

$$G_q = M + O(1), \quad M = \max_j \left\{ \frac{b_j}{b_{j-1}} \log |z_j| \right\}.$$

Let  $w$  be the monomial valuation with positive weights  $w_j = b_{j-1}/b_j$ . For convergent germs it is equivalently  $w(f) = \sigma(\log |f|, M)$ . One way to see this relative-type formula is to use the weighted polydiscs  $\{|z_j| \leq r^{w_j}\}$ , on which  $M \leq \log r$ . The Taylor series gives the bound from the least weighted degree; Cauchy's coefficient estimates show that no larger degree can satisfy the required uniform bound. The formula is valid for positive real weights, since only finitely many monomials have weighted degree below any fixed bound. Counting lattice points in  $\{\alpha \in \mathbb{R}_{\geq 0}^n : \sum_j w_j \alpha_j < m\}$  gives

$$(7.10) \quad \text{vol}(w) = \frac{1}{w_1 \cdots w_n} = b_n = \text{vol}(v).$$

We next prove  $w \geq v$  on  $R_{\text{an}}$ . If  $f \in I_m$ , then  $\log |f| \leq \log |I_m| + O(1)$  and hence  $\log |f| \leq G_{I_m, D} + O(1)$  near 0. The latter comparison can also be read directly from polynomial generators: shift their bounded-on- $D$  logarithmic norm by a constant to obtain an admissible negative representative. Whenever  $m$  divides  $k!$ , the graded property gives

$$\frac{1}{m} G_{I_m, D} \leq \frac{1}{k!} G_{I_{k!}, D} = q_k \leq q.$$

After shrinking to the domain used for  $G_q$ , still  $q \leq G_q$ , so

$$\log |f| \leq m G_q + O(1) = m M + O(1).$$

Therefore  $w(f) \geq m$ . For any nonzero nonunit convergent germ  $f$ , apply this to  $f^r \in I_{\lfloor rv(f) \rfloor}$  for all sufficiently large  $r$ . It yields

$$r w(f) \geq \lfloor rv(f) \rfloor.$$

Division by  $r$  and passage to the limit prove  $w(f) \geq v(f)$ . For units both values are zero.

Both valuations satisfy positive lower and finite upper ordinary order bounds, by (7.1) and positivity of the weights of  $w$ . Finite Taylor truncations therefore extend this inequality from convergent germs to all of  $\widehat{R}$ . Now (7.10) and lemma 7.5 give  $w = v$ . The coordinate functions were obtained holomorphically from theorem 1.1, so they are convergent; their invertible linear part makes them a regular system of parameters on the completion. This proves the stated formula for every formal series.  $\square$

*Remark 7.6* (Scope of the consequence). For a closed-point center and finite discrepancy, equality  $E_l(\mathfrak{a}_\bullet(v)) = A_R(v)$  with  $l < n$  is impossible: all  $b_j$  are positive and finite, so  $E_l < E_n \leq A_R(v)$ . Thus corollary 1.2 handles every

possible equality with a closed-point center under the stated finite-discrepancy hypothesis.

For a nonmaximal center  $\mathfrak{p}$ , one has  $v(\mathfrak{m}) = 0$ . The ambient primary power bounds and positive top-dimensional volume used in this section no longer apply. The next section instead restricts the graded ideals to a general section and lifts the resulting coordinates back to  $R$ . Localizing at  $\mathfrak{p}$  alone would not suffice: the desired parameters belong to the original ring, not just the local ring at the generic point of the center.

## 8. ARBITRARY CENTERS VIA MULTIPLIER-IDEAL LIFTING

We prove [corollary 1.3](#) from [theorem 1.1](#). The new issue is to recover parameters in the original ring after applying the analytic classification on a section. We restrict the valuation ideals, not the valuation: a valuation need not descend to the quotient by a linear section.

### 8.1. A bounded-value lemma.

**Lemma 8.1** (Finiteness of bounded sets of values). *Let  $v$  be a real valuation of the fraction field of a Noetherian domain  $R$ , nonnegative on  $R$ , with nonzero center  $\mathfrak{p}$ . For every finite  $B$ , the set*

$$\{v(f) : 0 \neq f \in R, v(f) \leq B\}$$

*is finite.*

*Proof.* Set  $T = R_{\mathfrak{p}}$  and  $\mathfrak{q} = \mathfrak{p}T$ . Elements of  $R \setminus \mathfrak{p}$  have value zero, so  $v$  is centered at the maximal ideal of  $T$  and is nonnegative on  $T$ . Since  $\mathfrak{q}$  is finitely generated,  $\beta = v(\mathfrak{q}) > 0$ . Choose  $N$  with  $N\beta > B$ . For every  $\gamma \in [0, B]$ , the ideal

$$P_{\gamma} = \{f \in T : v(f) \geq \gamma\}$$

contains  $\mathfrak{q}^N$ . Distinct realized values in  $[0, B]$  give distinct members of this nested family: if  $v(f) = \gamma < \gamma'$ , then  $f \in P_{\gamma} \setminus P_{\gamma'}$ . The ideals  $P_{\gamma}/\mathfrak{q}^N$  form a chain of submodules of the finite-length module  $T/\mathfrak{q}^N$ , hence only finitely many of them are distinct. This also bounds the realized values coming from  $R$ .  $\square$

We shall also use the discrepancy retraction inequality [[12](#), Corollary 5.4]. For any regular system of parameters  $z_1, \dots, z_n$  of  $R$ , let  $r_z(v)$  be the monomial retraction with these coordinate values. Then

$$(8.1) \quad \sum_{j=1}^n v(z_j) = A_R(r_z(v)) \leq A_R(v),$$

and equality holds if and only if  $v$  is monomial in these parameters. This includes zero coordinate values: the retraction is taken on the coordinate stratum containing the center. In particular every regular parameter has value at most  $A_R(v)$ .

**8.2. A primary graded family on a general section.** Write  $\mathfrak{a}_m = \mathfrak{a}_m(v)$  and  $A = A_R(v)$ . Choose a single very general linear section

$$S = R/(t_1, \dots, t_{n-l}) \simeq \mathbb{C}[[x'_1, \dots, x'_l]]$$

such that  $\mathfrak{p}S$  is primary to the maximal ideal  $\mathfrak{n}$  of  $S$  and, for every  $m$  and  $1 \leq j \leq l$ ,

$$e_j(\mathfrak{a}_m S) = \sigma_j(\mathfrak{a}_m).$$

These are the usual general-section formulas; one avoids the countably many exceptional sets for the countable family of ideals. The height hypothesis ensures that the intersection of the center with such a section is zero-dimensional. Put  $\mathfrak{b}_m = \mathfrak{a}_m S$ . The  $\mathfrak{b}_m$  are a graded family of primary ideals, without any assertion that they are valuation ideals.

Let  $\delta = v(\mathfrak{p}) > 0$ , and choose  $N$  with  $\mathfrak{n}^N \subseteq \mathfrak{p}S$ . Then

$$(8.2) \quad \mathfrak{n}^{N \lceil m/\delta \rceil} \subseteq \mathfrak{b}_m.$$

Indeed  $\mathfrak{p}^{N \lceil m/\delta \rceil} \subseteq \mathfrak{a}_m$ . The restriction theorem for multiplier ideals on regular excellent schemes, [12, Theorem A.1], applied successively along the linear flag, gives

$$\mathrm{lct}_S(\mathfrak{b}_m) \leq \mathrm{lct}_R(\mathfrak{a}_m) \leq \frac{A}{v(\mathfrak{a}_m)} \leq \frac{A}{m}.$$

Skoda's lower bound  $\mathrm{lct}_S(\mathfrak{b}_m) \geq 1/\mathrm{ord}_{\mathfrak{n}}(\mathfrak{b}_m)$  therefore yields

$$(8.3) \quad \mathfrak{b}_m \subseteq \mathfrak{n}^{\lceil m/A \rceil}.$$

These two bounds replace the ambient closed-point Izumi bounds.

Contract to the convergent ring  $S_{\mathrm{an}} = \mathbb{C}\{x'_1, \dots, x'_l\}$ :  $I_m = \mathfrak{b}_m \cap S_{\mathrm{an}}$ . For fixed  $m$ , put  $K = N \lceil m/\delta \rceil$ . Every element of  $\mathfrak{b}_m$  has the same class modulo  $\mathfrak{n}^K$  as a polynomial of degree less than  $K$  that still belongs to  $\mathfrak{b}_m$ . A vector-space basis of  $\mathfrak{b}_m/\mathfrak{n}^K$ , together with the degree- $K$  monomials, therefore gives polynomial generators. As in lemma 7.1, this proves  $I_m S = \mathfrak{b}_m$ , with the same multiplicities, thresholds, and multiplier ideals after completion. By the same common-section and volume-multiplicity argument as in section 7, the limits

$$b_j = \lim_{m \rightarrow \infty} \frac{e_j(I_m)}{m^j} \quad (1 \leq j \leq l), \quad b_0 = 1,$$

exist and form a positive, finite log-convex vector. Our hypothesis and the threshold bounds imply

$$(8.4) \quad A = \sum_{j=1}^l \frac{b_{j-1}}{b_j} \leq \mathrm{lct}(I_{\bullet}) \leq A.$$

On a fixed bounded hyperconvex domain in  $\mathbb{C}^l$ , form

$$q_k = \frac{1}{k!} G_{I_{k!}, D}, \quad q = \left( \sup_k q_k \right)^*.$$

The power bounds required in (7.6) hold with  $\rho = A^{-1}$  and  $C = N/\delta + N$ , since

$$N\lceil m/\delta \rceil \leq (N/\delta + N)m \quad (m \geq 1).$$

Equation (8.4) is exactly the extremality hypothesis (7.7) in dimension  $l$ . Thus proposition 7.3 applies and gives

$$(8.5) \quad c_0(q) = E_l(q) = A, \quad e_j(q) = b_j.$$

Apply theorem 1.1 in dimension  $l$ . There are holomorphic coordinates  $y_1, \dots, y_l$  on the section with

$$(8.6) \quad G_q = M + O(1), \quad M = \max_i \frac{\log |y_i|}{w_i}, \quad w_i = \frac{b_{i-1}}{b_i} > 0, \quad \sum_i w_i = A.$$

### 8.3. Multiplier ideals of the increasing realization.

**Lemma 8.2.** *For every  $\lambda > 0$ , in the convergent local ring of the section,*

$$\mathcal{J}(\lambda M) = \mathcal{J}(\lambda q) = \bigcup_k \mathcal{J}(I_{k!}^{\lambda/k!}).$$

Consequently, if  $\lambda < A + w_i$ , then for some  $m = k!$ ,  $y_i \in \mathcal{J}_S(\mathfrak{b}_m^{\lambda/m})$ .

*Proof.* By lemma 2.3 and (8.6),  $\mathcal{J}(\lambda M) = \mathcal{J}(\lambda q)$  and  $\mathcal{J}(\lambda q_k) = \mathcal{J}(I_{k!}^{\lambda/k!})$ . The sequence  $q_k$  is increasing on a common domain, so lemma 2.2 identifies the second ideal with the union in the statement. In particular this is an assertion about the full ideals, rather than just their zero sets or their unweighted integrability thresholds.

By lemma 2.4,  $c_0^{y_i}(M) = \sum_j w_j + w_i = A + w_i$ . For  $\lambda < A + w_i$  it follows that  $y_i \in \mathcal{J}(\lambda M)$ , hence  $y_i \in \mathcal{J}(I_{k!}^{\lambda/k!})$  for some finite index  $k$ . The exponent  $\lambda$  is fixed when this index is chosen; the index may change as  $\lambda$  approaches  $A + w_i$ . Finally, [12, Proposition 1.9] identifies the completion of this analytic multiplier ideal with  $\mathcal{J}_S(\mathfrak{b}_{k!}^{\lambda/k!})$ . This gives the required membership in the formal ring  $S$ .  $\square$

### 8.4. Coordinate lifting and discrepancy equality.

**Lemma 8.3** (A valuative bound for multiplier-ideal membership). *Let  $R$  be a regular excellent local ring over  $\mathbb{C}$ , let  $v$  have finite log discrepancy and a center on  $\text{Spec } R$ , and let  $I \neq 0$  be an ideal. If  $\lambda \geq 0$  and  $0 \neq F \in \mathcal{J}_R(I^\lambda)$ , then*

$$v(F) + A_R(v) \geq \lambda v(I).$$

*Proof.* Choose a log resolution  $\pi : Y \rightarrow \text{Spec } R$  of the ideal and of the divisor of  $F$ . Enlarge its normal-crossing divisor to include the exceptional locus. Write  $d_E$  for the coefficient of  $I\mathcal{O}_Y$  along a component  $E$ ,  $k_E$  for that of  $K_{Y/R}$ , and  $f_E = \text{ord}_E(F)$ . The defining formula for multiplier ideals implies

$$f_E + k_E \geq \lfloor \lambda d_E \rfloor, \quad f_E + A_R(\text{ord}_E) \geq \lambda d_E.$$

Let  $r(v)$  be the monomial retraction on this log-smooth pair, and write  $\alpha_E \geq 0$  for its weights on the components through the center of  $v$  on  $Y$ . The pullbacks of both  $I$  and  $F$  are monomial times units near that center, so

$$v(I) = \sum_E \alpha_E d_E, \quad v(F) = \sum_E \alpha_E f_E.$$

Moreover  $A_R(r(v)) = \sum_E \alpha_E A_R(\text{ord}_E) \leq A_R(v)$  by [12, Corollary 5.4]. Multiply the componentwise inequalities by  $\alpha_E$  and sum to obtain the result. Log resolutions and the multiplier-ideal formula in this argument apply to the formal rings used here by the framework of [12].  $\square$

Fix  $i$  and  $\lambda < A + w_i$ , with  $\lambda > 0$ . Choose  $m$  as in lemma 8.2. Applying the restriction theorem along the chosen flag gives

$$(8.7) \quad \mathcal{J}_S(\mathfrak{b}_m^{\lambda/m}) \subseteq \mathcal{J}_R(\mathfrak{a}_m^{\lambda/m})S.$$

The ideal on the right is the image of the ideal on  $R$ . Hence  $y_i$  has a lift  $F_{i,\lambda} \in R$  such that

$$F_{i,\lambda}|_S = y_i, \quad F_{i,\lambda} \in \mathcal{J}_R(\mathfrak{a}_m^{\lambda/m}).$$

Applying lemma 8.3 gives

$$(8.8) \quad v(F_{i,\lambda}) + A \geq \frac{\lambda}{m} v(\mathfrak{a}_m) \geq \lambda.$$

Only this non-strict estimate is needed; no limiting extension at the critical exponent is required.

Letting  $\lambda$  approach  $A + w_i$  from below shows

$$\sup\{v(F) : F|_S = y_i\} \geq w_i.$$

Every such lift is a regular parameter: any lifts of all the  $y_j$ , together with the  $t_j$ , have linearly independent images in  $\mathfrak{m}/\mathfrak{m}^2$ . By (8.1), its value is at most  $A$ . The set of possible lift values is nonempty and is a subset of the finite set in lemma 8.1 with  $B = A$ . Its supremum is therefore a maximum. Choose lifts  $F_i$  with

$$(8.9) \quad F_i|_S = y_i, \quad v(F_i) \geq w_i \quad (1 \leq i \leq l).$$

This attainment argument takes place after localizing only for the bounded-value lemma; all the actual lifts belong to  $R$ . It does not assume  $v(\mathfrak{m}) > 0$ , convergence of the lifts, or smoothness of the center.

Finally,  $F_1, \dots, F_l, t_1, \dots, t_{n-l}$  is a regular system of parameters of  $R$ . Indeed, modulo the span of the linear terms of the  $t_j$ , the classes of the  $F_i$  are the independent linear terms of the coordinates  $y_i$  on  $S$ . The full set therefore forms a basis of  $\mathfrak{m}/\mathfrak{m}^2$ , and the formal inverse function theorem applies. Apply (8.1):

$$(8.10) \quad A \geq \sum_{i=1}^l v(F_i) + \sum_{j=1}^{n-l} v(t_j) \geq \sum_{i=1}^l w_i = A.$$

All terms are nonnegative, so every inequality is an equality,  $v(F_i) = w_i$ , and  $v(t_j) = 0$ . The equality criterion in [12, Corollary 5.4] says that  $v$  equals its monomial retraction in this very system of parameters. Its center is therefore  $(F_1, \dots, F_l)$ , which has height  $l$  and regular quotient. Explicitly, every nonzero formal series can be written uniquely as

$$f = \sum_{\alpha \in \mathbb{N}^l} c_\alpha(t) F^\alpha, \quad c_\alpha(t) \in \mathbb{C}[[t_1, \dots, t_{n-l}]],$$

and monomiality gives

$$v(f) = \min_{c_\alpha(t) \neq 0} \sum_{i=1}^l \alpha_i w_i.$$

The positive weights ensure that the displayed minimum exists. It is positive exactly when  $c_0(t) = 0$ , proving the assertion about the center in the original ring. This proves [corollary 1.3](#).

*Remark 8.4* (What the extension requires). The new argument adds two published inputs to the analytic classification: the multiplier-ideal restriction theorem on regular excellent schemes and the equality criterion for the discrepancy of a monomial retraction. Both apply to formal power series rings. No regularity of the center or finite generation of its valuation ideals is assumed. The coordinate changes supplied by the restriction theorem belong to the formal ring. The convergence assertion available at a closed-point center in [corollary 1.2](#) comes instead from holomorphic extension and is not needed for the present conclusion.

## 9. EXAMPLES AND FURTHER QUESTIONS

*Example 9.1* (The original germ need not have bounded logarithmic error). Let  $M_a = \max_j a_j \log |z_j|$  with  $a_j > 0$ , and work near 0 where  $M_a < -1$ . Set

$$u = M_a - \sqrt{-M_a}.$$

The function  $t \mapsto t - \sqrt{-t}$  is increasing and convex, so  $u$  is psh. For every  $\varepsilon > 0$ , near 0 one has

$$(1 + \varepsilon)M_a \leq u \leq M_a.$$

Comparison and passage to  $\varepsilon \downarrow 0$  show that  $e_j(u) = e_j(M_a)$  and  $c_0(u) = c_0(M_a)$ . Thus  $u$  is an equality germ, but  $u - M_a$  is unbounded. In fact no holomorphic change of coordinates makes  $u$  a diagonal logarithmic maximum modulo bounded terms. If another such model existed, its coefficients would have the same ordered mixed-mass ratios. On a smooth curve transverse to both systems of coordinate hyperplanes, both logarithmic models have leading term  $a_1 \log |t| + O(1)$ , whereas  $u$  has the additional unbounded term  $-\sqrt{-a_1 \log |t| + O(1)}$ .

*Remark 9.2* (The meaning of general psh). The theorem concerns all germs in  $\mathcal{E}_{\text{loc}}$  with positive Lelong number; it does not presuppose an isolated singularity. Equality forces the greenification to have an isolated logarithmic

lower bound through [proposition 3.3](#). An unrestricted psh germ need not have a defined top Monge–Ampère mass. No classification is asserted here for truncated inequalities in positive-dimensional singularity loci.

*Remark 9.3* (Analytic and algebraic ingredients). The critical truncation, the Zhou reduction, and exact lifting are analytic. The hypersurface identity uses the graded ideals of the Zhou valuation and [\(4.5\)](#). Replacing that input with a direct Monge–Ampère proof would yield a proof formulated entirely in pluripotential and extension theory. The present manuscript makes no such replacement claim.

*Remark 9.4* (Truncated equality for arbitrary psh germs). The valuation consequence treats arbitrary centers because the graded ideals admit primary images on general sections, and their multiplier ideals satisfy a restriction theorem on the original formal ring. For a general psh germ with a positive-dimensional singularity locus, an analogous classification would require a separate formulation of the transverse masses and of the lifting step. [Theorem 1.1](#) concerns the full-dimensional invariant on  $\mathcal{E}_{\text{loc}}$  and does not by itself assert such a truncated classification. This distinction also separates the present statement from the Lelong-number-one theorem of [\[10\]](#).

#### DECLARATION ON THE USE OF ARTIFICIAL INTELLIGENCE

The author selected the research problem, proposed the analytic direction, and suggested the extension to arbitrary centers. The arguments and exposition were developed with OpenAI Codex (GPT-6). The mathematical content remains subject to the author’s verification; no independent verification or peer review is claimed for this version.

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